

Sri Lanka Institute of Information Technology



**Artificial Intelligence and Machine Learning - IT2011
Year 2 - Semester 1 - 2025**

STOCK MARKET PRICE PREDICTION USING MACHINE LEARNING

**Model Evaluation, Comparison, and Final Selection
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Model Performance Summary and Final Selection

- This section presents all models developed.
- Each model was trained, tuned, and evaluated using consistent data splits and performance metrics.

Table 1 – Summary of Developed Models and Performance Metrics

Model	BEST VERSION	R ² Score	MAE	RMSE	Accuracy (Up/Down)	F1	AUC
Random Forest Regressor	Baseline Model	0.981186	134.998	185.541	0.519	0.567	0.62
XGBoost Regressor	All-Features Variant	0.9724	177.3194	225.8072	0.8162	0.8245	0.89
LightGBM Regressor	Tuned GridSearchCV Model	0.9982	50.7800	167.3424	0.9982	0.981	0.990
SVR	Best RBF Kernel (after RandomizedSearchCV)	0.9626	0.037 (scaled)	0.0478	0.85 (scaled)	0.85 (scaled)	0.90 (scaled)
ANN	LSTM_Model (Best ANN Version)	0.8779	367.55	478.63	0.9869	NA	NA
LSTM	Variety 2 (Deeper LSTM)	-	2574.49	2855.80	0.4798	0.1077	0.48

Overall Model Ranking

- 1) **LightGBM Regressor**
- 2) **XGBoost Regressor**
- 3) **Random Forest Regressor**
- 4) ANN (Artificial Neural Network)
- 5) SVR (Support Vector Regressor)
- 6) LSTM (Long Short-Term Memory)

Ranking Criteria

- 1) R^2 (Coefficient of Determination)
 - How much of the variance in actual price is explained by the model.
 - Higher is better.
- 2) MAE (Mean Absolute Error)
 - Average magnitude of prediction errors.
 - Lower is better.
- 3) RMSE (Root Mean Squared Error)
 - Square root of average squared errors.
 - Lower is better.
- 4) Accuracy
 - Percentage of correctly predicted up/down movements.
 - Higher is better.
- 5) F1 Score
 - Balance between precision (correct positivity) and recall (capturing all positives)
 - Higher is better
- 6) AUC (Area Under Curve)
 - Probability that the model ranks a true positive higher than a false one.
 - Higher is better

Best Combination: High R^2 + Low MAE + Low RMSE + High Accuracy + High F1 + High ACU

Challenges, Performance, and Expected Behavior

1) Challenges Faced

- High Market Volatility
- Non-stationary Data
- Overfitting Risk
- Data Imbalance
- Hyperparameter Optimization:
- Feature Correlation
- Limited Features

2) Performance Overview

- Regression Metrics: Models were primarily evaluated using R^2 , MAE, and RMSE for price prediction.
- Classification Metrics: Used Accuracy, F1 Score, and AUC to assess next-day movement prediction (Up/Down).
- Cross-validation: TimeSeriesSplit ensured temporal consistency and prevented data leakage.
- Result Summary:
 - Ensemble models (LightGBM, Random Forest, XGBoost) achieved highest R^2 (~0.99) and lowest MAE.
 - SVR performed well on small samples but was slower for large datasets.
 - ANN showed moderate learning but required more epochs for stability.
 - LSTM effectively captured temporal dependencies but struggled with short-term daily predictions.

3) Expected Behavior of Models

MODEL	EXPECTED BEHAVIOR
Random Forest Regressor	Expected to perform consistently well in both regression and classification due to ensemble averaging and resistance to overfitting. Provides strong interpretability through feature importance.
XGBoost Regressor	Expected to achieve high predictive accuracy through gradient boosting and built-in regularization. Performs slightly better than Random Forest for complex non-linear interactions but requires careful hyperparameter tuning
LightGBM Regressor	Expected to deliver the best overall performance — faster training, high accuracy, and efficient handling of large feature sets. Particularly suitable for financial time-series forecasting.
SVR (Support Vector Regressor)	Expected to perform well on smaller or medium-sized datasets with clear margin separation. Handles non-linear patterns effectively but can become computationally heavy as data size increases.

ANN (Artificial Neural Network)	Expected to capture complex, non-linear feature relationships. Performs best with large datasets and proper hyperparameter optimization; may overfit small data if not regularized.
LSTM (Long Short-Term Memory)	Expected to excel at learning sequential and temporal dependencies in stock data. Performs better in long-term trend analysis than short-term next-day forecasts.

- Ensemble models (LightGBM, Random Forest ,XGBoost) were expected to lead due to robustness and interpretability.
- Deep learning models (ANN, LSTM) were expected to improve with more data and longer training.
- SVR was expected to show strong baseline regression but limited scalability.